University of Geneva

# Social Aid Chronicity in Geneva: A Project Proposal

(Empirical Project)

Alessandro Casorella
Master of Science in Economics

Supervised by Christophe Gaillac,
within the course *Advanced Topics in Economics*

May 2026

# 1 Introduction and Research Questions

Enacted in 2025, the *Loi sur l'Assistance Sociale et la Lutte contre la Précarité* (LASLP) aims to revitalize Geneva's social assistance system (Grand Conseil du Canton de Genève, 2023). The reform seeks to reduce the systematic controls carried out by the Hospice Général on beneficiaries, which would foster their autonomy and accelerate bureaucratic processes (Praz Dessimoz and Dergalenko, 2024).

The LASLP is a response to the increasing dependency on social assistance observed over the past fifteen years (Département de la cohésion sociale, 2019). However, it would be valuable to assess whether the shortcomings this law purports to address are truly institutional in nature. Does the worsening of welfare chronicity really reflect a failure of the old system? Or could it instead be driven by economic conditions or the labour market?

To address these questions, we evaluate the effectiveness of the former Geneva social assistance scheme: the *Loi sur l'Assistance Sociale Individuelle* (LIASI), enacted in 2007 (Grand Conseil du Canton de Genève, 2007). The LIASI represents a major overhaul of the cantonal social assistance system. It amends the *Loi sur l'Assistance Publique* of 1981 (LAP) by introducing conditional assistance procedures linked to beneficiaries' efforts toward professional rehabilitation (Grand Conseil du Canton de Genève, 1980). Job assessment internships were introduced, as well as sanctions in cases of non-compliance with reintegration efforts (Grand Conseil du Canton de Genève, 2007).

This paper aims to analyse the causal effect of the LIASI on different proxies of welfare dependency exit and labour market reintegration. If the latter indeed proved ineffective, it is likely that the LASLP addresses genuine institutional shortcomings (this, of course, does not constitute a causal inference, but merely a heuristic interpretation). We focus on two research questions:

- What was the causal effect of the LIASI on welfare dependency and on the professional reintegration of beneficiaries? This would inform us about the overall effectiveness of the law in the medium to long term.
- Did the LIASI have stronger positive effects on forward-looking individuals? It may be the case that individuals oriented toward the future, and therefore more sensitive to the anticipated consequences of potential sanctions, were more responsive to the transition toward a conditional assistance system. If treatment effects are found to be highly heterogeneous, this would suggest that the 2007 reform had only partial effectiveness, benefiting only certain types of individuals.

# 2 Literature Review

It is worth examining the contribution of the literature to the study of workfare, the mechanism of conditional social assistance linked to professional reintegration that underpins the LIASI.

According to Besley and Coate (1992), workfare acts as a tax on inactivity: it reduces the free-riding problem, the temptation to benefit from public assistance even when one does not need it. According to the two authors, this system therefore reduces dependency on social assistance only for a certain type of individual: those whose human and social capital is sufficiently high for them to potentially exit assistance, but who would prefer to benefit from it as long as possible. This paper thus highlights the heterogeneity of workfare effects stemming from the socio-economic segmentation of individuals, whereas our study of the LIASI will contribute to establishing the heterogeneity of its effects through a psychological segmentation (individuals who are more or less forward-looking).

We also consider, in this review, a report by Gachet and Muehlethaler (2023) on the chronicity of social assistance, conducted in 2023. This report, referring to the situation of the Canton of Geneva between 2012 and 2022, highlights an increase in the number of long-term recipients over time, explained largely by the precarisation of the labour market, a tightening of eligibility criteria for unemployment and disability insurance, and cohort effects. Through the exploitation of longitudinal data, this report provides an overview of the origins of worsening welfare dependency in the canton. However, no causal identification is provided.

Finally, we consider a study by Sansone and Zhu (2021) which, drawing on Australian administrative data, exploited Machine Learning methods to predict which socio-economic profiles are most at risk of chronic welfare dependency. This article highlights that, in the presence of strong outcome heterogeneity, with a sufficiently large sample and in a multivariate context, complex predictive algorithms such as Gradient Boosting or Random Forests are far more effective than linear predictors. Although this study likewise provides no causal design, it is one of the few papers to use Machine Learning methods to predict welfare dependency. Our analysis of the LIASI, by contrast, will exploit a causal design through Double Machine Learning methods to better study Swiss social assistance data.

# 3 Data Collection and Problem Setting

**Data Acquisition and Unit of Observation.** Our analysis draws on longitudinal data from the Swiss Household Panel (SHP), available in Stata format through the SWISSUbase research data repository, which has collected socio-demographic information from thousands of Swiss households since 1999 (FORS – Centre de compétences suisse en sciences sociales, 2024; Atlas of Longitudinal Datasets, 2024). The SHP is a nationally representative panel study, with samples drawn from the sampling frames of the Swiss Federal Statistical Office, and addresses the inherent attrition limitations of panel data through approximately two refreshment samples per decade (Atlas of Longitudinal Datasets, 2024). Although the SHP database comprises several files, we focus primarily on the two longfiles, which contain the consolidated panel data for individuals and households respectively. We prioritize the individual-level longfile, as the individual is the unit of observation in our study, and draw only a limited number of additional variables from the household-level file.

The individual longfile contains social, financial, demographic, health, and attitudinal data on 49,876 individuals over the period 1999–2023, across 974 variables. From the household longfile, we extract 10 variables, namely *canton*, our treatment variable, as well as proxies for the forward-looking score. These variables are originally recorded at the household level and are merged into the individual longfile using *year* and *idhous* as join keys.

**Sample Selection.** As our identification strategy relies on Double Machine Learning (DML), we do not filter observations by eligibility age for benefits or any other criterion: non-linearities and discontinuities in the relationship between our outcomes and covariates will be handled by the algorithm. All observations from the merged longfile are therefore retained, initially.

We drop observations with missing values on the outcome variables, while missing values on all remaining variables are imputed by the median for continuous variables and by the mode for categorical variables. However, variables with more than 50% missing values are excluded from the analysis.

After these steps, the analytical sample comprises 298,989 observations across 380 variables.

**Treatment Assignment.** In order to exploit the longitudinal nature of the data, we assign treatment through a binary variable taking the value of 1 for individuals residing in Geneva in 2006, just before the implementation of the LIASI, and 0 for all others, the key of our unit of observation being *idpers*. As the Geneva LIASI constitutes our treatment, treated individuals are those residing in Geneva, year by year. Relocations between Geneva and other cantons therefore create a non-compliance phenomenon between treatment and its assignment. However, we assume that the ITT, the effect of treatment assignment on outcomes, provides a close proxy for the treatment effect itself. This assumption is supported by two empirical observations:

- The number of non-compliers is negligible: among treated observations, only 219 person-year observations concern individuals recorded in another canton at least once during the panel, compared to 5,691 pure treated person-year observations, that is, those assigned to treatment in 2006 and never recorded outside Geneva. Non-compliers introduce a slight attenuation bias, meaning that the estimated ITT is a conservative lower bound of the true treatment effect.

- There is no evidence of spillover effects: an inspection of relocation dynamics between 2000 and 2010 reveals no substantial increase in out-migration from Geneva around the 2007 reform. Had such a pattern been observed, it would suggest that individuals were selectively fleeing the LIASI, implying a particular profile among out-migrants and introducing bias through the misclassification of treated individuals who, though assigned to treatment, were no longer effectively exposed to it.

**Outcomes and Forward-Looking Score.** We consider three outcomes. First, a welfare persistence score, defined as the share of years with positive welfare receipt over the observation window (2000–2006 pre-treatment, 2009–2015 post-treatment), capturing the mechanical effect of the treatment on benefit dependency. Second, an unemployment persistence score, constructed analogously, measuring the indirect effect of the treatment on the transition towards professional rehabilitation. Third, cumulative gross employment income, summed over 2000–2006 and 2010–2016 respectively, and transformed via an inverse hyperbolic sine to account for the mass of zeros and high values, capturing the material autonomy gained through treatment.

Outcomes are constructed as period-level aggregates that repeat across individual-year observations, which could artificially inflate the statistical validity of inference. Collapsing to a single pre/post value per individual, however, would sacrifice the panel structure; we therefore cluster standard errors at the individual level (*idpers*) to account for within-person correlation instead (though this only partially addresses the issue). It is also worth noting that the models only exploit observations corresponding to the relevant years, that is, 2000–2006 and 2009–2015 for the persistence scores, and 2000–2006 and 2010–2016 for employment income; observations outside these windows are not considered.

We also construct a forward-looking score as a weighted composite of 3 pre-treatment indicators: third pillar investment, frequency of saving habits, and housing ownership status. These proxies, though limited in number, capture the degree to which individuals orient their financial decisions towards the future, and will be used in the heterogeneous ITT analysis.

# 4 Empirical Strategy

**Identification Assumptions.** The models presented in the following sections aim to compare, through a difference-in-differences specification, the treatment-assigned group to the control group before and after the reform, the transition year being 2007, with the goal of identifying the ITT. We therefore examine, for each outcome, the plausibility of parallel trends between the two assignment groups. We first verify unconditional parallel trends, then conditional parallel trends across the most extreme quantiles of the forward-looking score, as these are necessary to identify heterogeneous effects in the Causal Random Forest specification. It should be noted that, as outcomes are cross-sectional aggregates, parallel trends are assessed on the raw variables from which they are constructed, namely annual welfare amounts, unemployment benefits, and gross employment income. For the first two, the raw variables are substantially more volatile than the final outcomes, which are based on the presence or absence of benefits rather than their amounts; the parallel trends test is therefore conservative for these outcomes.

Across all outcomes and in both scenarios, we observe not only a lack of pre-treatment parallel trends between the two assignment groups, but also that the control group itself is not always pure. Conditioning the parallel trends analysis on additional covariates could have helped distinguish whether the divergence stems from compositional bias, which a sufficiently rich set of controls could resolve, but the volatility of the treatment group outcomes suggests that creating covariate-specific counterfactual subgroups would lead to excessive statistical noise. The diverging trends are not encouraging, but proposed resolutions are discussed in the following sections.

The common support condition, however, is satisfied. We verify this by estimating a global propensity score, that is, the conditional probability of residing in Geneva given a set of key covariates (age, household income, forward-looking score, working status, education level, residence permit, and sex) on the pre-treatment sample, and plotting its density across both groups. The distributions overlap across the entire support, meaning that valid counterfactuals exist for every treated individual. Moreover, the control group is clustered towards low probabilities of treatment, while the treatment distribution is shifted to the right, confirming the presence of selection bias and the necessity of flexible covariate adjustment.

Before proceeding, we assume that, conditionally on the covariates available in the dataset, treatment assignment is independent of potential outcomes. This assumption is plausible given that our dataset comprises 380 socio-demographic, financial, and behavioural covariates; however, differences in outcomes between Geneva residents and the rest of Switzerland may also reflect structural or institutional factors not captured by our micro-panel. Robustness checks related to the parallel trends assumption, developed in subsequent sections, will serve to test for potential violations of the CIA and to adapt our models accordingly.

**Baseline Specification.** We first estimate a naive difference-in-differences model, with clustered standard errors at the individual level:

$$Y_{it} = \beta_1 \cdot (\text{treated}_i \times \text{post}_t) + \beta_2 \cdot \text{age}_{it} + \beta_3 \cdot \text{idispy}_i^{\text{pre}} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{1}$$

where $Y_{it}$ denotes the outcome of interest, $\text{treated}_i \times \text{post}_t$ is the DiD interaction term (the coefficient $\beta_1$ being the ITT on the treatment-assigned group), $\text{idispy}_i^{\text{pre}}$ is household income measured in 2006, $\mathbf{X}_{it}$ is a vector of additional controls, and $\alpha_i$ and $\gamma_t$ are individual and year fixed effects (individual fixed effects control for time-invariant unobserved heterogeneity, and year fixed effects absorb common temporal shocks).

This model is unable to handle confounding selection given the limited number of covariates integrated heuristically. Moreover, household income (*idispy*) is a pre-treatment variable; to avoid introducing post-treatment values, which risk acting as colliders, we retain only its 2006 value as a covariate. Incorporating the full panel structure with a gap around 2007 would have broken the longitudinal continuity of the variable.

Within Double Lasso, that comes next, introducing individual and temporal fixed effects directly risks overfitting the data. We therefore replace them with manually computed within-transformations: individual fixed effects are absorbed by subtracting, for each variable, the individual-level mean computed across time at constant $i$, and temporal fixed effects are absorbed by subtracting the time-level mean computed across individuals at constant $t$.

**Double Lasso.** To estimate the effect of treatment assignment on each outcome, we employ a Double Lasso estimator within the same DiD framework.

First, we regress the outcome $Y$ on the full set of covariates using Lasso, selecting the regularisation parameter $\lambda$ via cross-validation, and record in $\hat{S}_Y$ the set of covariates with non-zero coefficients. Second, we regress the treatment $D$ on the full set of covariates using Lasso, and record the corresponding selected set $\hat{S}_D$.

The union $\hat{S}_Y \cup \hat{S}_D$ ensures that variables relevant for either outcome prediction or treatment assignment are retained. This step mitigates omitted variable bias that could arise from regularisation shrinking relevant covariates to zero in finite samples.

Finally, we regress $Y$ on $D$ and the variables in $\hat{S}_Y \cup \hat{S}_D$ via OLS. Inference is conducted using cross-fitting, to reduce overfitting.

Note: for variables that may be affected by treatment, we construct pre-treatment versions of these covariates, as done for *idispy* in the baseline specification.

**Causal Random Forest.** On one hand, the Causal Random Forest is used within a DiD framework to estimate the average ITT by introducing non-linear learners (which the Double Lasso does not capture); on the other hand, it is used to estimate heterogeneous ITT effects based on the forward-looking score.

We estimate a causal forest per outcome and obtain, for each individual, a predicted ITT $T(X_i)$ (the CRF first estimates nuisance functions for both the outcome and the treatment via separate random forests, then recovers individual ITT estimates from the resulting residuals), through sample splitting. This reduces overfitting, as prediction and estimation are performed on separate subsamples (at the cost of higher variance).

We then exploit these individual-level predictions to recover aggregate features of the ITT distribution. That way, we estimate the average ITT and a heterogeneity parameter by regressing a proxy of the ITT on $T(X_i)$.

Next, we construct Sorted Group Average Treatment Effects (GATES) by partitioning individuals according to the quantiles of $T(X_i)$ and estimating the average ITT within each group.

Finally, we conduct a Classification Analysis by comparing observable characteristics across groups defined by $T(X_i)$, focusing on the forward-looking score. That way, we can assess whether forward-looking individuals systematically exhibit larger ITT effects.

**Credibility Checks.** To assess the validity of our strategy, we implement placebo DiD specifications in pre-treatment periods within a DML framework.

We examine whether estimated placebo ITT and heterogeneous ITT are statistically distinguishable from zero. Null results are not a definitive validation, but they may suggest parallel trends hold.

We also assess the stability of outcomes over time, conditional on covariates.

If these validity checks are not satisfied, the DiD-based estimates may not be causally interpretable. In that case, alternative approaches explicitly accounting for divergent pre-trends, such as synthetic control methods, would be more suitable.

**Final note.** The overall identification strategy of this analysis is based on the baseline DiD specification, while machine learning methods are introduced as extensions designed to relax the structural assumptions of the problem (targeting the prediction of both outcomes and treatment assignment via MSE loss functions and cross-validation to prevent overfitting). The Double Lasso framework is used to address confounder selection and strengthen the credibility of the CIA, whereas the Causal Random Forest adds an additional layer by introducing non-linear learners and enabling the analysis of ITT heterogeneity.

# References


Atlas of Longitudinal Datasets (2024). Swiss household panel (SHP). Atlas of Longitudinal Datasets. Accessed May 2024.

Besley, T. and Coate, S. (1992). Workfare versus welfare: Incentive arguments for work requirements in poverty-alleviation programs. *American Economic Review*, 82(1):249–261.

Département de la cohésion sociale (2019). *Rapport de la commission chargée de la révision de la loi cantonale sur l'insertion et l'aide sociale individuelle (LIASI).* Rapport technique, République et Canton de Genève.

FORS – Centre de compétences suisse en sciences sociales (2024). Swiss household panel – living in switzerland. SWISSUbase. Version 14.4, Ref. 6097.

Gachet, N. and Muehlethaler, I. (2023). *Etude sur le phénomène de la longue durée à l'aide sociale – 2ème partie.* Rapport interne, Hospice général, Pôle Recherche et Développement.

Grand Conseil du Canton de Genève (1980). *Loi sur l'assistance publique (LAP) du 19 septembre 1980.*

Grand Conseil du Canton de Genève (2007). *Loi sur l'insertion et l'aide sociale individuelle (LIASI) du 22 mars 2007 (J 4 04).*

Grand Conseil du Canton de Genève (2023). *Loi sur l'assistance sociale et la lutte contre la précarité (LASLP) (13119).*

Praz Dessimoz, Y. and Dergalenko, N. (2024). Genève mise sur confiance, responsabilité et autonomie. *L'Action sociale, Hospice général Genève.*

Sansone, D. and Zhu, A. (2021). Using machine learning to create an early warning system for welfare recipients. IZA Discussion Paper 14377, IZA Institute of Labor Economics.

# Appendix A: Technical Details for Double Lasso

As we already mentioned in the report, we want to absorb individual and temporal fixed effects without introducing them explicitly in the Lasso regression. Thus, we apply manual within-transformations prior to estimation. For any variable $V_{it}$:

$$\ddot{V}_{it} := V_{it} - \bar{V}_{i\cdot} - \bar{V}_{\cdot t} + \bar{V}_{\cdot\cdot}, \tag{2}$$

where $\bar{V}_{i\cdot}$ is the individual-level mean across time at constant $i$, $\bar{V}_{\cdot t}$ is the time-level mean across individuals at constant $t$, and $\bar{V}_{\cdot\cdot}$ is the overall mean. We apply this transformation to all variables before entering them into the Lasso algorithm. That way, for each variable (whether outcome, treatment, or covariate), individual-specific and time-specific means are removed, so that the demeaned variables $\ddot{V}_{it}$ retain only within-individual, within-period variation, implicitly absorbing both fixed effects without penalizing them through the Lasso.

**Covariate Selection.** The outcome Lasso minimizes:

$$\mathcal{L}_Y(\beta) = \frac{1}{n}\sum_{i,t}\left(\ddot{Y}_{it} - \ddot{\mathbf{X}}_{it}'\beta\right)^2 + \lambda_Y\|\beta\|_1, \tag{3}$$

yielding selected set $\hat{S}_Y$, the set of covariates with non-zero coefficients in the outcome regression. The treatment Lasso minimizes:

$$\mathcal{L}_D(\delta) = \frac{1}{n}\sum_{i,t}\left(\ddot{D}_{it} - \ddot{\mathbf{X}}_{it}'\delta\right)^2 + \lambda_D\|\delta\|_1, \tag{4}$$

yielding selected set $\hat{S}_D$, where $\ddot{D}_{it}$ is the within-transformed DiD interaction term. In both regressions, $\ddot{\mathbf{X}}_{it}$ contains all demeaned covariates that vary across both dimensions.

The Lasso steps solely aim to select which controls to retain and do not carry the causal DiD structure. The regularization parameters $\lambda_Y$ and $\lambda_D$ are selected via $K$-fold cross-validation with $K \leq 5$, given our moderate sample size.

The union $\hat{S}_Y \cup \hat{S}_D$ ensures that variables relevant for either outcome prediction or treatment assignment are retained. Including $\hat{S}_D$ is particularly important: variables correlated with $\ddot{D}_{it}$ but whose coefficient on $\ddot{Y}_{it}$ is small may be incorrectly shrunk to zero by the outcome Lasso, yet omitting them would introduce a confounding bias in the final OLS estimate of $\hat{\beta}_1$.

The DiD structure is fully restored only in the final OLS step, where $\ddot{Y}_{it}$ is regressed on $\ddot{D}_{it}$ and the covariates indexed by $\hat{S}_Y \cup \hat{S}_D$, recovering the ITT estimate $\hat{\beta}_1$.

**Estimation and cross-fitting.** Naively introducing the controls selected by Lasso into the final OLS could yield a biased ITT estimator, as many relevant controls may be omitted due to the regularization parameter. We therefore combine double selection with cross-fitting: the sample is partitioned into $K$ folds, and for each fold $k$, the two Lasso regressions are estimated on the complementary subsample while the final OLS is evaluated on fold $k$, ensuring that the selection and estimation steps are performed on independent subsamples and preventing overfitting from contaminating inference on $\hat{\beta}_1$.

# Appendix B: Technical Details for Causal Random Forest

**Nuisance Parameter Estimation.** We first estimate two separate random forests in order to control for the effect of $\ddot{\mathbf{X}}_{it}$ on both the outcome and the treatment variable. The first forest predicts the within-transformed outcome $\ddot{Y}_{it}$, which gives residuals defined as $\tilde{Y}_{it} := \ddot{Y}_{it} - \hat{m}(\ddot{\mathbf{X}}_{it})$. The second forest predicts the within-transformed DiD interaction term $\ddot{D}_{it}$, where $D_{it} = \text{treated}_i \times \text{post}_t$, and produces residuals $\tilde{D}_{it} := \ddot{D}_{it} - \hat{p}(\ddot{\mathbf{X}}_{it})$, with $\hat{p}(\cdot)$ denoting the estimated propensity score conditional on the observed covariates.

In both forests, splits are selected by minimizing the MSE:

$$\mathcal{L}_{\text{MSE}}(\ell) = \frac{1}{|\ell|} \sum_{i \in \ell} \left(V_{it} - \bar{V}_\ell\right)^2, \tag{5}$$

where $\ell$ denotes a leaf, $\bar{V}_\ell$ is the average within the leaf, and $V_{it} \in \{\ddot{Y}_{it}, \ddot{D}_{it}\}$.

For each tree, 80% of the sample is randomly drawn without replacement for training, while the remaining 20% is used as out-of-bag (OOB) data to evaluate prediction error. The number of trees $B$ starts at 100 and is increased until the OOB error stabilizes. The minimum leaf size is fixed at $k = 5$, and the number of candidate variables considered at each split follows the standard rule $m = \lfloor p/3 \rfloor$.

Both nuisance forests are estimated using cross-fitting. The sample is partitioned into $K$ folds and, for each fold $k$, the forests are trained on the complementary subsample before residuals are predicted on fold $k$. Residuals are therefore always predicted on observations that are excluded from the estimation sample.

**Honest Causal Forest.** The causal forest is then estimated on the cross-fitted residuals $(\tilde{Y}_{it}, \tilde{D}_{it})$, while keeping the same hyperparameters $B$, $k$, and $m$. Each tree is built on a subsample that is itself divided into two disjoint sets, denoted $\mathcal{J}$ and $\mathcal{I}$. The first set is used to determine the tree structure, whereas treatment effects are estimated only on the second set. This honest procedure avoids using the same observations both to construct the partition and to estimate the leaf-level treatment effects.

At each split, the algorithm searches for the partition that maximizes differences in treatment effects across the two resulting branches. For a candidate split producing a left branch $G$ and a right branch $D$:

$$\Delta = \frac{N_G \cdot N_D}{N_G + N_D} \left(\hat{\tau}_G - \hat{\tau}_D\right)^2, \tag{6}$$

where the local treatment effect in each branch is estimated as:

$$\hat{\tau}_b = \frac{\sum_{i \in b \cap \mathcal{J}} \tilde{Y}_{it} \tilde{D}_{it}}{\sum_{i \in b \cap \mathcal{J}} \tilde{D}_{it}^2}. \tag{7}$$

Since the estimation relies on residualized variables, the splitting procedure focuses on variation in treatment effects after accounting for observed confounding factors. In addition, $\alpha$-regularity is imposed with $\alpha = 0.05$, meaning that each split must leave at least 5% of observations on both sides.

**Heterogeneity Analysis.** Given that each individual is characterized by a set of covariates $X_i$, the forest produces a predicted ITT $T(X_i)$ for every individual by averaging the leaf-level estimates across all trees. These individual predictions can then be aggregated using inverse propensity-score weights in order to recover the average ITT.

To study treatment effect heterogeneity, individuals are partitioned into high and low groups according to the forward-looking score. Sorted Group Average Treatment Effects (GATES) are then estimated within each subgroup. Comparing these two GATES indicates whether the LIASI generates larger ITT effects for individuals displaying stronger forward-looking behavior.

# Appendix C: Conceptual Note on Synthetic Control

Should the robustness checks described at the end of the empirical strategy fail, we would conclude that the instability of parallel trends does not stem from compositional differences in the micro-characteristics of the Genevan population relative to the rest of Switzerland. In this scenario, the diverging trends would reflect structural or institutional differences between Geneva and the rest of Switzerland that cannot be reduced to the micro-variables available in our panel.

In that case, replacing our DML specification with a Synthetic Control Method (SCM) would be worth considering. Taking Geneva as the treated unit and drawing from the rest of Switzerland as a donor pool, we would use pre-treatment values of the outcomes as well as key predictors from the panel to construct a synthetic canton whose pre-treatment outcome trajectory closely matches that of Geneva.

This approach would yield a control unit with pre-treatment outcomes parallel to those of the treatment group by construction. In return, it would not guarantee the purity of the control group and would require aggregating the individual-level panel to the canton level, entailing a substantial loss of micro-level richness — both in terms of available covariates and in the ability to study treatment effect heterogeneity across individuals.

## Paragraph on AI Disclosure

I used generative AI tools (ChatGPT and Gemini) for assistance with R code syntax, to deepen my understanding of the course's theoretical concepts, and to further develop econometric notions with which I had limited familiarity. AI was also used to assist with specific LaTeX formatting and syntax.